given the bivariate function P(Ha, Hb), how to transform its variables in order to transform its integration over a triangular region defied by the following limits for its two variables: Ha=1..H and Hb=1..Ha into an integration over a rectangular region?

To transform an integration over a triangular region into one over a rectangular region, you need a coordinate transformation that decouples the limits of the variables. In your case, the dependency is $1 \leq H_b \leq H_a$.

The most common technique for this is a variation of the **Duffy Transformation**, which maps a triangle to a square (or rectangle) by using a scaling factor.

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1. The Original Integral

The integral over the triangular region is defined as:

$$\int_1^H \int_1^{H_a} P(H_a, H_b)\, d H_b\, d H_a$$

The region of integration is a right triangle in the (H_a, H_b) plane with vertices at $(1,1)$, $(H,1)$, and (H,H).

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2. Defining the Transformation

To map this to a rectangular domain where the limits are constants, we define two new variables, u and v.

We want u to represent the normalized range of H_a and v to represent the relative position of H_b within its allowed range $[1, H_a]$.

1. **Transform H_a:**

   Let u scale linearly from 1 to H.

   $$H_a = 1 + u(H-1), \text{where } u \in [0,1]$$

   (Alternatively, you can simply keep $H_a = u$ where $u \in [1, H]$ if you don't need a unit square).

2. **Transform H_b:**

   Since H_b starts at 1 and ends at H_a, we define v such that it scales the "width" of the triangle at any given H_a:

$$H_b = 1 + v(H_a - 1)$$

   Substituting the expression for H_a:

$$H_b = 1 + v(u(H-1)), \text{where } v \in [0,1]$$

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3. Calculating the Jacobian

When changing variables in a multiple integral, you must multiply by the absolute value of the **Jacobian determinant** $|J|$ to account for the stretching of the area element.

Given:

- $H_a(u,v) = 1 + u(H-1)$
- $H_b(u,v) = 1 + uv(H-1)$

The Jacobian matrix is:

$$J = \begin{pmatrix} \frac{\partial H_a}{\partial u} & \frac{\partial H_a}{\partial v} \\ \frac{\partial H_b}{\partial u} & \frac{\partial H_b}{\partial v} \end{pmatrix} = \begin{pmatrix} H-1 & 0 \\ v(H-1) & u(H-1) \end{pmatrix}$$

The determinant is:

$$|J| = (H-1) \cdot u(H-1) - 0 = u(H-1)^2$$

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4. The Transformed Integral

The integral now over the rectangular (unit square) region $u \in [0,1], v \in [0,1]$ is:

$$\int_0^1 \int_0^1 P(1 + u(H-1), 1 + uv(H-1)) \cdot u(H-1)^2 \, dv \, du$$

Summary of the Mapping:

Original Variable	Transformation Equation	New Range
H_a	$1+u(H-1)$	$u \in [0,1]$
H_b	$1+uv(H-1)$	$v \in [0,1]$
Area Element	$dH_b dH_a \to u(H-1)^2 du dv$	-

This transformation is particularly useful for numerical integration (like Gaussian Quadrature), as it allows you to evaluate the integral over a standard $[0,1]\times[0,1]$ grid.

Does this specific transformation suit the numerical method you're planning to use, or are you looking for a simpler shift for symbolic integration?

> given the bivariate function P(Ha, Hb), how to transform its variables in order to transform its integration over a triangular region defied by the following limits for its two variables: Ha=H..Hmax and Hb=Ha..Hmax into an integration over a rectangular region?

To transform an integration over a triangular region into a rectangular one, we use a mapping (often called a **Duffy transformation**) that decouples the limits of the inner integral from the outer variable.

In your case, the region is defined by:

- $H \le H_a \le H_{max}$
- $H_a \le H_b \le H_{max}$

This corresponds to an **upper triangle** in the (H_a, H_b) plane with vertices at (H,H), (H,H_{max}), and (H_{max},H_{max}).

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1. Defining the Transformation

We want to map the triangular region to a unit square where both new variables, u and v, range from 0 to 1.

Let $L=H_{max}-H$ be the total length of the interval.

1. **Transform H_a (The outer variable):**

We map H_a linearly over the interval $[H, H_{max}]$ using u:

$$H_a = H + u(H_{max} - H)$$

When $u=0, H_a = H$. When $u=1, H_a = H_{max}$.

2. **Transform H_b (The inner variable):**

   Since the limits of H_b depend on H_a (H_b goes from H_a to H_{max}), we define v as the relative position within that remaining "slice":

$$H_b = H_a + v(H_{max} - H_a)$$

   By substituting the expression for H_a, we get:

$$H_b = [H + u(H_{max} - H)] + v[H_{max} - (H + u(H_{max} - H))]$$

$$H_b = H + u(H_{max} - H) + v(1-u)(H_{max} - H)$$

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2. Calculating the Jacobian

To change variables in the double integral, we must calculate the determinant of the Jacobian matrix, J, to account for the change in the area element $dH_b dH_a$.

Given $L = H_{max} - H$:

- $H_a(u, v) = H + uL$
- $H_b(u, v) = H + uL + vL(1-u)$

The partial derivatives are:

- $\frac{\partial H_a}{\partial u} = L$
- $\frac{\partial H_a}{\partial v} = 0$
- $\frac{\partial H_b}{\partial u} = L - vL = L(1-v)$
- $\frac{\partial H_b}{\partial v} = L(1-u)$

The Jacobian determinant is:

$$|J| = \left| \frac{\partial H_a}{\partial u}\frac{\partial H_b}{\partial v} - \frac{\partial H_a}{\partial v}\frac{\partial H_b}{\partial u} \right| = |L \cdot L(1-u) - 0| = L^2(1-u)$$

3. The Transformed Integral

The original integral over the triangle:

$$\int_{H}^{H_{max}} \int_{H_a}^{H_{max}} P(H_a, H_b)\, d H_b\, d H_a$$

Becomes the following integral over a **rectangular region** (the unit square):

$$\int_0^1 \int_0^1 P(H+u L, H+u L+v L(1-u)) \cdot L^2 (1-u)\, d v\, d u$$

Summary Table

Variable	Transformation	Range
H_a	$H+u(H_{max}-H)$	$u \in [0,1]$
H_b	$H_a+v(H_{max}-H_a)$	$v \in [0,1]$
Differential	$d H_b\, d H_a \to (H_{max}-H)^2 (1-u)$	-

This transformation is very effective for numerical integration (like Gaussian Quadrature) because it eliminates the variable boundary, allowing you to use a standard rectangular grid of evaluation points.

Are you planning to implement this in a specific programming language or numerical solver?

given the bivariate function P(Ha, Hb), how to transform its variables in order to transform its integration over a triangular region defied by the following limits for its two variables: Ha=H..Hmax and Hb=H..Ha into an integration over a rectangular region?

This specific triangular region is slightly different from your previous one. Here, H_b starts at a fixed base H and grows until it hits the diagonal $H_b = H_a$. This forms a right triangle with vertices at (H, H), (H_{max}, H), and (H_{max}, H_{max}).

To map this to a rectangular domain (the unit square where $u, v \in [0,1]$), we can use the following transformation.

1. Defining the Mapping

We want to decouple the variables so that the limits of v no longer depend on u. Let $L = H_{max} - H$ represent the total range.

- **For H_a (The independent limit):**

  We scale H_a linearly across the full range $[H, H_{max}]$.

  $$H_a = H + u(H_{max} - H) = H + uL$$

  (When $u=0, H_a = H$; when $u=1, H_a = H_{max}$)

- **For H_b (The dependent limit):**

  Since H_b ranges from H to H_a, we define v as the fraction of that specific distance:

  $$H_b = H + v(H_a - H)$$

  Substituting the expression for H_a:

  $$H_b = H + v(H + uL - H) = H + uvL$$

  (When $v=0, H_b = H$; when $v=1, H_b = H_a$)

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2. Calculating the Jacobian

Since we are changing the coordinate system, we must scale the area element $dH_b dH_a$ by the determinant of the Jacobian matrix, J.

The partial derivatives are:

- $\frac{\partial H_a}{\partial u} = L$
- $\frac{\partial H_a}{\partial v} = 0$
- $\frac{\partial H_b}{\partial u} = vL$
- $\frac{\partial H_b}{\partial v} = uL$

The Jacobian determinant is:

$$|J| = \left| \left(\frac{\partial H_a}{\partial u} \cdot \frac{\partial H_b}{\partial v} \right) - \left(\frac{\partial H_a}{\partial v} \cdot \frac{\partial H_b}{\partial u} \right) \right| = \left| (L \cdot u L) - (0 \cdot v L) \right| = u L^2$$

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3. The Transformed Integral

The original integral:

$$\int_{H}^{H_{max}} \int_{H}^{H_a} P(H_a, H_b) \, d H_b \, d H_a$$

Becomes the rectangular integral:

$$\int_{0}^{1} \int_{0}^{1} P(H + u L, H + u v L) \cdot u L^2 \, d v \, d u$$

Quick Reference Table

Variable	Transformation	Range
H_a	$H + u(H_{max} - H)$	$u \in [0,1]$
H_b	$H + u v (H_{max} - H)$	$v \in [0,1]$
Area Factor	$(H_{max} - H)^2 u$	-

Why this works

By defining H_b as a function of $u v$, you are essentially “stretching” the triangle into a square. As u gets smaller (moving toward the left of the triangle), the range of H_b shrinks. The u in the Jacobian accounts for this compression of area near the vertex (H, H).

Does this version align better with the data structure or the physics of the problem you’re modeling?

do not scale the domain to the unit square but to a square limited by H and Hmax

Got it—you want to maintain the physical range of your variables rather than normalizing them. To transform the triangle $H \le H_a \le H_{max}, H \le H_b \le H_a$ into a square defined by $u \in [H, H_{max}]$ and $v \in [H, H_{max}]$, we can use a linear interpolation approach.

1. Defining the Transformation

We will map H_a directly to u, and then define H_b as a function of both u and v so that its range "stretches" to fill the square.

Let $L = H_{max} - H$.

1. **Map H_a:**

   We set H_a to be exactly u.

   $$H_a = u$$

   Since $u \in [H, H_{max}]$, H_a naturally covers its original range.

2. **Map H_b:**

   We need H_b to go from H to u as v goes from H to H_{max}. Using linear interpolation:

   $$H_b = H + \frac{(u - H)}{(H_{max} - H)}(v - H)$$

   Check the limits:

   - When $v = H \Longrightarrow H_b = H$ (The bottom edge).
   - When $v = H_{max} \Longrightarrow H_b = H + (u - H) = u$ (The diagonal edge).

2. Calculating the Jacobian

Even though we aren't scaling to a unit square, we are still distorting the area, so we need the Jacobian determinant $|J|$ for the transformation $(H_a, H_b) \to (u, v)$.

The partial derivatives are:

- $\frac{\partial H_a}{\partial u} = 1$
- $\frac{\partial H_a}{\partial v} = 0$
- $\frac{\partial H_b}{\partial u} = \frac{v - H}{L}$

- $\frac{\partial H_b}{\partial v} = \frac{u - H}{L}$

The Jacobian determinant is:

$$|J| = \left| \left(1 \cdot \frac{u - H}{L}\right) - \left(0 \cdot \frac{v - H}{L}\right) \right| = \frac{u - H}{H_{max} - H}$$

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3. The Transformed Integral

The original triangular integration:

$$\int_{H}^{H_{max}} \int_{H}^{H_a} P(H_a, H_b)\, d H_b\, d H_a$$

Becomes the rectangular integration over $u, v \in [H, H_{max}]$:

$$\int_{H}^{H_{max}} \int_{H}^{H_{max}} P\left(u, H + \frac{(u - H)(v - H)}{H_{max} - H}\right) \cdot \frac{u - H}{H_{max} - H}\, d v\, d u$$

Summary of the Mapping

Variable	Transformation Equation	Range
H_a	u	$u \in [H, H_{max}]$
H_b	$H + \frac{(u - H)(v - H)}{H_{max} - H}$	$v \in [H, H_{max}]$
Scaling Factor	$\frac{u - H}{H_{max} - H}$	-

> **Note:** The factor $\frac{u - H}{H_{max} - H}$ essentially acts as a weight that gets smaller as u approaches the lower bound H, which makes sense because the "width" of the triangle is narrowing at that point.

Is this for a simulation where you need to keep the variables in their original units for easier debugging?